Proof. The proof framework follows from approximate Newton (second order method) literature [134, 135, 136, 137, 138, 139, 140, 141, 104, 105, 107, 110, 111, 142, 143, 144, 145].

Following from Lemma D.47, we know the Hessian of the loss function is positive definite.

Following from Lemma D.48, we know the Hessian of L is Lipschitz.

Following Section 9 in [105], by running Algorithm 4, we complete the proof. $\square$

We remark that ω denotes the exponent of matrix multiplication (i.e., n^ω is the time of multiplying an $n \times n$ matrix with another $n \times n$ matrix). The most naive algorithm gives $\omega = 3$. Currently, the state-of-the-art algorithm gives $\omega = 2.373$.